

# ISHVA CHINNASAMY

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## PROFESSIONAL SUMMARY

Data Scientist and AI/ML Engineer with an M.Sc. in Artificial Intelligence (CGPA 8.2) and proven experience deploying end-to-end AI systems, ETL pipelines, and BI dashboards. Built production-grade applications spanning medical AI (skin disease classification, brain tumor detection), AI security (MCP proxy gateway), and full-stack ML deployment using Flask, FastAPI, and Docker. Seeking roles in Data Science, ML Engineering, or Data Analytics.

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## EXPERIENCE

### Data Analyst Intern • Gravity AI

Dec 2025 – May 2026

- Built Python-based ETL pipelines to ingest, clean, and transform employment and labor market datasets, improving pipeline efficiency by 25%.
- Performed EDA on employment trends using Pandas and NumPy, surfacing actionable insights for product and strategy teams.
- Implemented automated data quality checks, achieving 98%+ data accuracy across all datasets.
- Developed Tableau and Power BI dashboards translating complex employment data into business intelligence for cross-functional stakeholders.
- Collaborated with product and engineering teams to translate business requirements into data solutions, contributing to 30% faster feature development cycles.

### Software Intern • Acmegrade

Jul 2024 – Dec 2024

- Assisted in development and testing of ERP modules; identified and reported 50+ bugs, contributing to 15% improvement in process efficiency.
  - Applied data analysis and requirements-gathering in cross-functional meetings to inform design decisions, reducing project development time by 20%.
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## PROJECTS

### Skin Scan AI (DermaCon-IN) – AI Dermatology Screening System

Jan 2025 – May 2026

**Stack:** Python (Flask, PyTorch), React 18 (Vite), Node.js (Express), MySQL, Docker Compose, Swin Transformer, CBM | [github.com/Ishva24/Skin-Scan](https://github.com/Ishva24/Skin-Scan)

- Built a full-stack AI skin disease screening system classifying 20+ conditions with confidence scoring and GradCAM/CBM explainability, featuring safety-aware triage for high-risk predictions.
- Implemented advanced model ensemble (Swin Transformer + Concept Bottleneck Model) for interpretable, clinically-aligned predictions.
- Integrated end-to-end doctor discovery and appointment booking workflow for Tamil Nadu dermatologists with real-time search, filtering, and verified reviews.

### Guardian-MCP – Zero-Trust AI Security Gateway

2026

**Stack:** TypeScript, Node.js, JWT (RS256), Docker Compose, SSE | [github.com/Ishva24/Guardian-MCP](https://github.com/Ishva24/Guardian-MCP)

- Built an enterprise-grade security middleware proxy for the Model Context Protocol (MCP), solving the unprotected attack surface when AI agents connect to production databases and APIs.
- Implemented a 6-stage security pipeline: asymmetric JWT auth (RS256), role-based tool scopes, semantic argument inspection (anti-injection, path traversal, null-byte), hot-reloadable policy engine, and real-time SSE audit dashboard.
- Enables organizations to enforce least-privilege access for LLM agents without requiring code changes to upstream agents or downstream servers.

## Brain Tumor Detection Using MRI Scans

Nov 2024 – Dec 2024

**Stack:** Python, TensorFlow/Keras, CNN, OpenCV | [github.com/Ishva24/Brain\\_tumor\\_detection](https://github.com/Ishva24/Brain_tumor_detection)

- Built a CNN-based deep learning model achieving 95% classification accuracy on MRI scans, reducing diagnostic turnaround time by 40%.
- Applied image preprocessing, data augmentation, and architecture optimization for imbalanced medical imaging datasets.

## Diabetes Prediction Model

Mar 2024 – May 2024

**Stack:** Python, scikit-learn, Random Forest, Pandas | [github.com/Ishva24/Diabetes\\_Prediction](https://github.com/Ishva24/Diabetes_Prediction)

- Achieved 85% prediction accuracy (Random Forest); implemented full ML pipeline with feature engineering, cross-validation, and ROC-AUC evaluation to enable early detection of high-risk individuals.

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## TECHNICAL SKILLS

**Machine Learning & AI:** scikit-learn, TensorFlow, Keras, PyTorch, CNN, Swin Transformer, XGBoost, Gradient Boosting, GradCAM, Concept Bottleneck Models, computer vision, feature engineering, hyperparameter tuning

**Data Analysis & BI:** SQL, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI, Streamlit, Excel (Advanced), EDA, KPI Dashboards, A/B Testing

**ML Engineering & Deployment:** Flask, FastAPI, REST APIs, Docker, Docker Compose, MLOps, inference optimization, OpenAPI, JWT (RS256)

**Data Engineering:** ETL pipelines, data cleaning, data quality frameworks, MySQL, Knex.js, data pipeline automation

**AI Security:** MCP protocol, zero-trust architecture, semantic argument inspection, role-based access control, real-time audit logging

**Programming & Tools:** Python, TypeScript, JavaScript, SQL, Java, C++ | GitHub, Jupyter, Google Colab, VS Code, Postman, Docker

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## EDUCATION

### Master of Science – Artificial Intelligence

May 2026

Karunya University, Coimbatore | CGPA: 8.2

### Bachelor of Science – Computer Science

May 2024

Karunya University, Coimbatore | CGPA: 8.75

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## CERTIFICATIONS

- IT Automation with Python – Google / Tata Strive, Coursera (2024)
- Fundamentals of AI on Microsoft Azure – Microsoft (2024)
- Data Analytics – Google / Coursera (2024)